

Phase-shift classification for integer tetration

A proof of the 21-value conjecture, its even-base refinement,
and the related 28-value modular classification

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Abstract

Integer power towers become extraordinarily large, but the first decimal digit that changes between successive heights has a small and rigid eventual dynamics. Marco Ripà's phase-shift conjecture, stated in the appendix of his 2025 paper and in OEIS A376842, predicts exactly 21 possible phase words when the cycle is anchored at the first height of permanently constant congruence speed. We give a self-contained elementary proof under the height-one convention explicitly recorded in OEIS A371074. The proof separates the 2-adic and 5-adic valuations of consecutive tower differences. Their competition determines whether the phase is always 5, follows multiplication in $\mathbb{F}_5^\times$, or belongs to a balanced family $a = 1 + 10^v c$. The balanced family accounts for precisely four additional words. We also prove the conjectured 14-word restriction for even bases, the 28-word classification at the alternative anchor of OEIS A376446, and natural density $1/900$ for each of the four words arising from balanced valuations. Explicit onset formulas, an unbounded sharp-onset family, a polynomial-time computation argument, and independently cross-checked exact integer programs accompany the proofs. A source audit identifies a height-zero indexing ambiguity and an apparent erroneous value at base 21; neither is used as a substitute for proving the conjecture.

Status of this document. This is a proof-bearing research draft, not a refereed publication or a formally verified development. The universal claims below rest on the written arguments, not on a finite search. The consulted sources still state the target as a conjecture; no earlier proof of the complete classification was located in the targeted literature check. That search does not establish priority. Existing congruence-speed results are explicitly credited and rederived where needed.

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1 The selected problem and its provenance

Fix an integer base $a > 1$ and define the right-associated power tower

$$T_0 = 1, \quad T_{b+1} = a^{T_b} \quad (b \geq 0). \quad (1.1)$$

Thus $T_b = a \uparrow\uparrow b$ for positive integer b . This article concerns integer tetration and modular arithmetic, not an analytic interpolation in the height. Higher up-arrow expressions often supply enormous tower heights, but nothing in the proof requires writing those heights or towers in decimal.

The selected problem is an explicit, finite-valued conjecture rather than a newly invented question. Ripà asked a 23-value version on MathOverflow in October 2024 [5]. The appendix of the 2025 paper gives the sharper 21-value version and an additional 14-value restriction for even bases [4]. The OEIS A376842 entry, revision 39 dated 30 March 2026, still labels the 21-value statement a conjecture [6]. We address that formulation. The older 23-value set contains two extra rotations, 1397 and 1793, which do not occur at the intended anchor.

The general stabilization of trailing digits and formulas for their rate of stabilization are established topics: see Ripà's 2020 and 2021 papers and the 2022 paper with Onnis [1, 2, 3]. Those results should not be confused with the phase-word classification. The present proof derives the particular valuation identities it uses, then supplies the missing classification and its anchoring argument.

1.1 The observable

Put

$$D_b = T_{b+1} - T_b > 0 \quad (b \geq 0), \quad k_b = \min\{v_2(D_b), v_5(D_b)\} \quad (b \geq 1). \quad (1.2)$$

For a nonzero integer n , $v_p(n)$ is the largest $e \geq 0$ with $p^e \mid n$. We also write $v_{10}(n) = \min\{v_2(n), v_5(n)\}$, although 10 is not prime. The number k_b counts the trailing decimal positions in which T_b and T_{b+1} agree, allowing leading zeroes. Their first differing position is $k_b + 1$, counted from the right.

Definition 1.1 (Phase and speed). For $b \geq 1$, define the phase digit by

$$s_b \equiv -\frac{D_b}{10^{k_b}} \pmod{10}, \quad 1 \leq s_b \leq 9. \quad (1.3)$$

Define the congruence speeds by

$$V_1 = k_1, \quad V_b = k_b - k_{b-1} \quad (b \geq 2). \quad (1.4)$$

When these speeds are eventually constant, write v for the eventual value and

$$B = \min\{b \geq 1 : V_j = v \text{ for every } j \geq b\}. \quad (1.5)$$

The word $\text{APS}(a)$ is the four digits $s_B s_{B+1} s_{B+2} s_{B+3}$, reduced to two digits when the two halves agree, and then to one digit when those two digits agree. Words are written as integers, as in the OEIS convention.

The negative sign in (1.3) is intentional: the original digit minus the corresponding next-height digit is being measured. Since the lower k_b digits agree, reducing $(T_b - T_{b+1})/10^{k_b}$ modulo 10 gives exactly that digit difference, including any borrow automatically.

1.2 A convention that changes the answer

The convention $V_1 = k_1$ is not optional bookkeeping. It is explicitly stated in OEIS A371074: the number of matching trailing digits of a and a^a is the congruence speed at height one, with leading zeroes included [8]. It also makes $k_b = \sum_{j=1}^b V_j$, as in Equation (1) of Ripà–Onnis [3].

Although $T_0 = 1$ is useful for algebra, we *do not* define $V_1 = k_1 - v_{10}(a - 1)$. That alternative shifts the starting point for some cycles. In particular it would replace $\text{APS}(901) = 19$ by 91. Section 13 records the ambiguity in the source formulations and checks it against their explicit examples. Below, $k_0^* = 0$ will mean only the artificial initial total used in (1.4); it is not an assertion about $v_{10}(D_0)$.

1.3 Examples before the proof

For $a = 3$, the first towers are $3, 27, 3^{27}, \dots$. One obtains $k_b = b - 1$, $V_1 = 0$, $V_b = 1$ for $b \geq 2$, and $B = 2$. The phases from height two are $4, 6, 4, 6, \dots$, so $\text{APS}(3) = 46$.

For $a = 57$, $k_b = 3b$ and $B = 1$. The successive phases are $2, 6, 8, 4, 2, 6, 8, 4, \dots$. Thus a three-digit stabilization speed coexists with a four-step phase period: $\text{APS}(57) = 2684$. The speed and the phase period are different invariants.

2 Main classification

Define the following sets of words:

$$\mathcal{E} = \{2, 4, 6, 8, 28, 46, 64, 82, 2486, 2684, 4268, 4862, 6248, 6842, 8426, 8624\}, \quad (2.1)$$

$$\mathcal{U} = \{9, 19, 3971, 7931\}. \quad (2.2)$$

Theorem 2.1 (The 21-value classification). *Let $a > 1$ be an integer not divisible by 10, and use Definition 1.1. The speeds become permanently constant at a finite height B , with $v > 0$. The phases are periodic from that very height, with least period dividing 4, and*

$$\text{APS}(a) \in \mathcal{E} \cup \{5\} \cup \mathcal{U}. \quad (2.3)$$

Every one of these 21 words occurs. More precisely, exactly one of the following alternatives holds at every height $b \geq B$:

- (i) $v_2(D_b) < v_5(D_b)$, and $s_b = 5$.
- (ii) $v_5(D_b) < v_2(D_b)$, and s_b is the even lift of a geometric progression in $\mathbb{F}_5^\times$.
- (iii) $a = 1 + 10^v c$ with $v \geq 2$ and $\gcd(c, 10) = 1$. In this case $B = 2$, $k_b = (b + 1)v$, and

$$s_b \equiv -c^{b+1} \pmod{10} \quad (b \geq 1). \quad (2.4)$$

In alternative (iii), residues $c \equiv 1, 3, 7, 9 \pmod{10}$ give, respectively, the words 9, 3971, 7931, 19.

Theorem 2.2 (The even-base refinement). *For even $a > 1$ with $5 \nmid a$, the exact range is*

$$\{2, 4, 6, 8, 28, 46, 64, 82, 2486, 4268, 4862, 6248, 6842, 8426\}. \quad (2.5)$$

In particular, the two even-digit words 2684 and 8624 occur for odd bases but never for even bases at the minimal-speed anchor.

The important qualification is “from that very height.” Merely proving that the phases eventually become periodic would not settle the conjecture: $\text{APS}(a)$ is anchored at the *first* permanently constant speed. The valuation comparison in Section 5 closes that gap.

3 Elementary valuation tools

The basic identity connecting adjacent differences is

$$D_b = T_b(a^{D_{b-1}} - 1) \quad (b \geq 1). \quad (3.1)$$

Indeed, $T_{b+1}/T_b = a^{T_b - T_{b-1}} = a^{D_{b-1}}$. This identity is an equality of ordinary integers; no infinite tower or p -adic analytic continuation is being assumed.

Lemma 3.1 (The 5-adic lifting calculation). *If $x = 1 + 5^r u$ with $r \geq 1$ and $5 \nmid u$, then for every positive integer n ,*

$$v_5(x^n - 1) = r + v_5(n), \quad \frac{x^n - 1}{5^{r+v_5(n)}} \equiv u \frac{n}{5^{v_5(n)}} \pmod{5}. \quad (3.2)$$

Proof. For an exponent w coprime to 5, the binomial theorem gives $(1 + 5^r u)^w \equiv 1 + w5^r u \pmod{5^{r+1}}$. Raising a number $1 + 5^j z$ to its fifth power gives $1 + 5^{j+1} z'$ with $z' \equiv z \pmod{5}$: the terms of binomial degree at least two are divisible by 5^{j+2} . Write $n = 5^e w$, apply this fifth-power step e times, and then the coprime exponent calculation. The displayed residue is nonzero, proving the exact valuation as well as the congruence. $\square$

Lemma 3.2 (The 2-adic lifting calculation). *For odd $a > 1$ put*

$$A = v_2(a - 1), \quad t = v_2(a - 1) + v_2(a + 1) - 1 = \max\{v_2(a - 1), v_2(a + 1)\}. \quad (3.3)$$

Then $t \geq 2$. For positive even n ,

$$v_2(a^n - 1) = t + v_2(n). \quad (3.4)$$

Proof. Exactly one of $a - 1, a + 1$ is divisible by 4, and the other has valuation one. For odd w , the factorizations of $a^w - 1$ and $a^w + 1$ show that their 2-adic valuations equal those of $a - 1$ and $a + 1$, respectively. If $n = 2^e w$ with $e \geq 1$, factor the successive differences of squares. After the first factorization, each extra factor is $a^{2^j w} + 1$ with $j \geq 1$, which is congruent to 2 modulo 8. Hence the valuation is $v_2(a - 1) + v_2(a + 1) + (e - 1) = t + e$. $\square$

For $5 \nmid a$, set

$$o = \text{ord}_5(a) \in \{1, 2, 4\}, \quad r = v_5(a^o - 1) \geq 1. \quad (3.5)$$

The value of r is $v_5(a - 1)$, $v_5(a + 1)$, or $v_5(a^2 + 1)$ according as $o = 1, 2, 4$. In the latter two cases the other factor of $a^o - 1$ is a 5-adic unit. Whenever $o \mid n$, Lemma 3.1 yields

$$v_5(a^n - 1) = r + v_5(n). \quad (3.6)$$

If $o \nmid n$, this valuation is zero.

4 Exact local valuations of tower differences

Write $x_b = v_2(D_b)$ and $y_b = v_5(D_b)$.

Proposition 4.1 (Odd bases coprime to 5). *For odd $a > 1$ with $5 \nmid a$, and every $b \geq 1$,*

$$x_b = A + bt, \quad y_b = (b + \varepsilon)r, \quad (4.1)$$

where

$$\varepsilon = \begin{cases} 1, & o = 1, \\ 0, & o = 2, \\ 0, & o = 4 \text{ and } a \equiv 1 \pmod{4}, \\ -1, & o = 4 \text{ and } a \equiv 3 \pmod{4}. \end{cases} \quad (4.2)$$

In particular, $k_b = \min\{A + bt, (b + \varepsilon)r\}$ exactly, not just asymptotically.

Proof. All D_b are even and T_b is odd. Start with $v_2(D_0) = A$ and apply (3.1) and Lemma 3.2; each step increases the valuation by t .

For the 5-adic part, when $o = 1$, $v_5(D_0) = v_5(a - 1) = r$, and (3.6) applies immediately and at every subsequent step. When $o = 2$, D_0 is even and is not divisible by 5, so $y_1 = r$; each later step adds r . When $o = 4$ and $a \equiv 1 \pmod{4}$, $4 \mid D_0$ and again $y_1 = r$.

Finally suppose $o = 4$ and $a \equiv 3 \pmod{4}$. Then $D_0 \equiv 2 \pmod{4}$, so $y_1 = 0$. Every odd-base tower T_b with $b \geq 1$ is congruent to a modulo 4: its exponent is odd. Consequently $4 \mid D_b$ for $b \geq 1$. Thus $y_2 = r$, and all subsequent steps add r . These four initial conditions give the four lines of (4.2). $\square$

Proposition 4.2 (Even bases). *For even a with $5 \nmid a$, put $e = v_2(a)$. For all $b \geq 1$,*

$$x_b = eT_{b-1}, \quad y_b = \max\{0, (b + \varepsilon)r\}, \quad (4.3)$$

where

$$\varepsilon = \begin{cases} 1, & o = 1, \\ -1, & o = 2, \\ -1, & o = 4 \text{ and } 4 \mid a, \\ -2, & o = 4 \text{ and } a \equiv 2 \pmod{4}. \end{cases} \quad (4.4)$$

Moreover, $x_b > y_b$ for every $b \geq 2$.

Proof. The second factor in (3.1) is odd, so $x_b = v_2(T_b) = eT_{b-1}$. If $o = 1$, the 5-adic calculation starts with $v_5(D_0) = r$. If $o = 2$, D_0 is odd, so $y_1 = 0$; D_1 is even, so $y_2 = r$. For $o = 4$, D_0 is odd, and $D_1 = T_2 - a \equiv -a \pmod{4}$ because $4 \mid T_2$. If $4 \mid a$ this gives $y_2 = r$; otherwise $y_2 = 0$ and $y_3 = r$, since $4 \mid D_2$. Thereafter each step adds r .

It remains to prove strict dominance, rather than merely eventual dominance. For $o = 1$, $a \geq 5^r + 1 > 3r$, so $x_2 = ea > 3r = y_2$. For $o = 2$, $a \geq 5^r - 1 > r = y_2$. For $o = 4$, the elementary inequality $5^a > a^2 + 1$ for $a \geq 2$ implies $r < a$, so $x_2 > r \geq y_2$. The inequality $5^a > a^2 + 1$ follows by induction from $a = 2$.

Also $T_{j+1} = a^{T_j} \geq 2T_j$, since $2^n \geq 2n$ for every positive integer n . Therefore $x_b \geq 2^{b-2}x_2$. In the first case $3 \cdot 2^{b-2} \geq b + 1$; in the other cases $2^{b-2} \geq b - 1$. The displayed strict inequalities persist for all $b \geq 2$. $\square$

Proposition 4.3 (Bases ending in 5). *For odd a divisible by 5, put $f = v_5(a)$ and retain A, t from (3.3). Then*

$$x_b = A + bt, \quad y_b = fT_{b-1} \quad (b \geq 1). \quad (4.5)$$

If $a \geq 15$, $y_b > x_b$ for all $b \geq 2$. If $a = 5$, this strict inequality holds for $b \geq 3$, and

$$k_1 = 1, \quad k_2 = 5, \quad k_b = 2b + 2 \quad (b \geq 3), \quad B = 4. \quad (4.6)$$

Proof. The 2-adic calculation is unchanged. The factor $a^{D_{b-1}} - 1$ in (3.1) is prime to 5, giving the formula for y_b . For $a \geq 15$ we have $a > 3t \geq A + 2t$: for $t = 2, 3$ this is immediate; for $t \geq 4$ use $a \geq 2^t - 1 > 3t$. Thus $y_2 = fa > x_2$. The same doubling argument as above preserves strict dominance over the linear function x_b . For $a = 5$, direct substitution gives $k_1 = 1$, $k_2 = 5$, and $y_3 = 3125 > x_3 = 8$; doubling then preserves $y_b > x_b$ for all $b \geq 3$, giving the displayed k_b values. The speeds are 1, 4, 3, 2, 2, $\dots$, proving $B = 4$. $\square$

5 Why the minimal speed anchor is already safe

For nonunit bases, the preceding propositions immediately give exact onsets:

$$B = \begin{cases} 2, & a \text{ even, } o = 1, e \geq 2r, \\ 3, & a \text{ even, } o = 1, e < 2r, \\ 3, & a \text{ even, } o = 4, a \equiv 2 \pmod{4}, \\ 2, & a \text{ even in the remaining cases.} \end{cases} \quad (5.1)$$

Indeed, in the first two lines $k_1 = \min(e, 2r)$ and $k_b = (b+1)r$ for $b \geq 2$. The other lines follow from their zero-valued initial 5-adic profiles. None has $B = 1$.

For $5 \mid a$, $a \geq 15$, the onset is

$$B = \begin{cases} 2, & k_1 = A + t, \\ 3, & k_1 < A + t, \end{cases} \quad k_1 = \min(A + t, f). \quad (5.2)$$

Together with $B(5) = 4$, these formulas place the anchor strictly inside the 2-adically dominant regime. No tie remains at the anchor.

For odd bases coprime to 5 we need to understand the minimum of two affine functions. This small point is central to the argument.

Lemma 5.1 (No isolated valuation tie at the anchor). *For the profiles in Proposition 4.1, let $v = \min(t, r)$. The speeds are eventually v . If $t \neq r$, the line with smaller slope is strictly lower at every integer $b \geq B$. If $t = r$, the two lines are either strictly ordered everywhere or identical everywhere.*

Proof. Suppose first $t > r$. Set $\delta_b = x_b - y_b = A - \varepsilon r + b(t - r)$, which is strictly increasing. For $b \geq 2$, direct subtraction of the two possible minima shows

$$V_b = r \iff \delta_{b-1} \geq 0. \quad (5.3)$$

If $\delta_{b-1} < 0$ and the smaller branch changes during the step, then $V_b = r - \delta_{b-1} > r$; if it does not change, $V_b = t > r$. Consequently $B \geq 2$ implies $\delta_B > 0$, and the inequality only strengthens. If $B = 1$, then $k_1 = r$. A tie there would require $x_1 = A + t = r$, impossible since $A \geq 1$ and $t > r$. So this case is strict too.

If $t < r$, exchange the two lines: set $\delta_b = y_b - x_b = \varepsilon r - A + b(r - t)$. The analogue of (5.3) is $V_b = t$ if and only if $\delta_{b-1} \geq 0$. The same proof gives strictness when $B \geq 2$. In fact $B = 1$ cannot occur in this case: if the lower-slope line is already minimal at height one, then $k_1 = A + t > t$; otherwise some later increment is not t .

Finally, if $t = r$ the difference $x_b - y_b = A - \varepsilon t$ is constant. $\square$

Identical lines require $A = \varepsilon t$. Since $A > 0$ and $\varepsilon \in \{-1, 0, 1\}$, this is equivalent to

$$\varepsilon = 1, \quad A = t = r. \quad (5.4)$$

Thus $a \equiv 1 \pmod{20}$ and $v_2(a-1) = v_5(a-1) = v \geq 2$. Equivalently, $a = 1 + 10^v c$ with $\gcd(c, 10) = 1$. These are exactly the balanced bases. For them $k_1 = 2v$ and $V_b = v$ for $b \geq 2$, so $B = 2$.

5.1 Closed onset formula for odd units

The proof gives a compact exact formula useful in computation. If $t = r$, then $B = 1$ when $k_1 = t$, and $B = 2$ otherwise. If $t \neq r$, define

$$(C, q) = \begin{cases} (A - \varepsilon r, t - r), & t > r, \\ (\varepsilon r - A, r - t), & t < r. \end{cases} \quad (5.5)$$

Here $q > 0$ and $\delta_b = C + bq$. Then

$$B = \begin{cases} 1, & k_1 = v \text{ and } C + q \geq 0, \\ \max\{2, 1 + \lceil -C/q \rceil\}, & \text{otherwise.} \end{cases} \quad (5.6)$$

This formula does not infer an onset from a finite run of equal speeds; it proves that every later speed agrees.

6 The phase dynamics in the three regimes

6.1 The 2-adic regime

If $x_b < y_b$, then $k_b = x_b$ and $D_b/10^{k_b}$ is odd and divisible by 5. Its residue modulo 10 is therefore 5, and so is its negative. Hence $s_b = 5$. The anchor lemma and the nonunit calculations show that this conclusion holds for *all* $b \geq B$ whenever this is the eventual regime.

6.2 The 5-adic regime: a four-element group

Suppose $y_b < x_b$ for all $b \geq B$. The base is prime to 5 and the constant speed is r . Every s_b is even and nonzero modulo 5. Let

$$u = \frac{a^0 - 1}{5^r}, \quad C_5 = \begin{cases} a^a \pmod{5}, & a \text{ odd,} \\ 1, & a \text{ even,} \end{cases} \quad \lambda = C_5 u o^{-1} 2^{-r} \in \mathbb{F}_5^\times. \quad (6.1)$$

All inverses in this formula are taken in $\mathbb{F}_5$, so they exist.

For $b \geq B$, we have $o \mid D_b$. For odd bases this follows from $4 \mid D_b$ at every positive height; for even bases $B \geq 2$ and $4 \mid D_b$ at those heights. Also $T_{b+1} = a^{T_b} \equiv C_5 \pmod{5}$: $T_b \equiv a \pmod{4}$ for odd bases and $T_b \equiv 0 \pmod{4}$ for even bases at the relevant heights.

Apply Lemma 3.1 to $a^{D_b} = (a^o)^{D_b/o}$ and use $D_{b+1} = T_{b+1}(a^{D_b} - 1)$. Since $k_{b+1} = k_b + r$, division by 10^{k_b+r} gives

$$\frac{D_{b+1}}{10^{k_{b+1}}} \equiv C_5 u o^{-1} 2^{-r} \frac{D_b}{10^{k_b}} \pmod{5}, \quad s_{b+1} \equiv \lambda s_b \pmod{5}. \quad (6.2)$$

The factor 2^{-r} is essential: the normalization removes powers of 10, not just powers of 5.

Let $\ell : \mathbb{F}_5^\times \rightarrow \{2, 4, 6, 8\}$ be the unique even representative modulo 10; explicitly $\ell(1) = 6$, $\ell(2) = 2$, $\ell(3) = 8$, and $\ell(4) = 4$. We have

$$s_{B+j} = \ell(s_B \lambda^j \pmod{5}). \quad (6.3)$$

The multiplicative group $\mathbb{F}_5^\times$ has order 4. Multipliers 1, 4, 2, 3 have orders 1, 2, 4, 4, respectively. As the initial nonzero residue varies, the resulting words are exactly the 16 members of $\mathcal{E}$ in (2.1). This proves both the period bound and the complete *allowable* even-digit list. Actual witnesses are supplied below.

6.3 Balanced valuations: the four exceptional words

Lemma 6.1 (A decimal first-order lifting identity). *Let $a = 1 + 10^v c$, $v \geq 2$, $\gcd(c, 10) = 1$. Then for every $b \geq 0$,*

$$D_b \equiv 10^{(b+1)v} c^{b+1} \pmod{10^{(b+1)v+1}}. \quad (6.4)$$

Proof. The valuation identities already show $v_2(D_b) = v_5(D_b) = (b+1)v$, including $b = 0$. The congruence starts with $D_0 = 10^v c$. Expand

$$a^{D_b} - 1 = D_b 10^v c + \sum_{j=2}^{D_b} \binom{D_b}{j} 10^{jv} c^j. \quad (6.5)$$

For $p = 2, 5$, the integer identity $j \binom{D_b}{j} = D_b \binom{D_b-1}{j-1}$ gives

$$v_p \binom{D_b}{j} \geq v_p(D_b) - v_p(j).$$

Every term with $j \geq 2$ therefore has p -adic valuation at least

$$(b+2)v + (j-1)v - v_p(j) \geq (b+2)v + 1.$$

The last inequality holds because $v \geq 2$ and $v_p(j) \leq j-1$. Thus all nonlinear terms disappear modulo $10^{(b+2)v+1}$. Multiply by $T_{b+1} \equiv 1 \pmod{10}$ in (3.1), divide by $10^{(b+2)v}$, and conclude that the normalized leading residue is multiplied by c at each step. Induction proves (6.4). $\square$

Since $B = 2$, the four possible words start with $-c^3$, not with $-c^2$. They are

$c \pmod{10}$	first four phase digits	reduced word	example base
1	9, 9, 9, 9	9	101
3	3, 9, 7, 1	3971	301
7	7, 9, 3, 1	7931	701
9	1, 9, 1, 9	19	901

The anchor, the sign convention, and the use of base 10 all matter in this table. In particular, arbitrary rotations of these words need not be attainable at B .

Proof of Theorem 2.1, except for the explicit witnesses. Propositions 4.1–4.3 give eventual positive speed. Section 5 proves that precisely one of strict 2-adic dominance, strict 5-adic dominance, or identical affine valuations holds from the minimal anchor onward. The three calculations in this section give the word 5, one of the 16 words of $\mathcal{E}$, or one of the four words of $\mathcal{U}$, respectively. In each case the phase period divides 4. The witness table in Section 8 proves that none of the 21 possibilities is superfluous. $\square$

7 Why even bases allow only fourteen words

The finite-group argument alone permits all 16 even-digit words. The even-base restriction comes from the initial residue and cannot be deduced just from $B \leq 3$.

For even a , $C_5 = 1$, so $\lambda = u o^{-1} 2^{-r}$. The first phase residue at the minimal anchor satisfies the following exact table, entirely in $\mathbb{F}_5$:

Condition	additional condition	B	$s_B \pmod{5}$
$o = 1$	$e \geq 2r$	2	$-\lambda^3$
$o = 1$	$e < 2r$	3	$-\lambda^4$
$o = 2$		2	3λ
$o = 4$	$4 \mid a, a \equiv 2 \pmod{5}$	2	λ
$o = 4$	$4 \mid a, a \equiv 3 \pmod{5}$	2	2λ
$o = 4$	$a \equiv 2 \pmod{4}$	3	3λ

To verify the first two rows, when $o = 1$ the normalized 5-adic residue obeys $D_b/5^{(b+1)r} \equiv u^{b+1} \pmod{5}$. Hence, once $k_b = (b+1)r$, $s_b \equiv -\lambda^{b+1}$. Formula (5.1) gives the two values of B .

In the next three rows, $B = 2$, $T_2 \equiv 1 \pmod{5}$, and $D_1 \equiv 1 - a \pmod{5}$. The lifting calculation gives $s_2 \equiv \lambda(a-1)$. This is 3λ when $a \equiv 4 \pmod{5}$, λ when $a \equiv 2 \pmod{5}$, and 2λ when $a \equiv 3 \pmod{5}$. In the last row, $T_2 \equiv -1 \pmod{5}$ while $T_3 \equiv 1 \pmod{5}$, so $D_2 \equiv 2 \pmod{5}$. Thus $s_3 \equiv -2\lambda = 3\lambda$.

For $\lambda = 2$, the table permits initial residues 1, 2, 4, but not 3. The missing start gives exactly the word 8624. For $\lambda = 3$, it permits 1, 3, 4, but not 2; the missing start gives 2684. For $\lambda = 1$ all four constant phases remain possible, and for $\lambda = 4$ all four period-two words remain possible. This proves the 14-word upper bound. The even-base witnesses in the next section prove equality, completing Theorem 2.2.

8 Explicit witnesses and exact certificates

The following table gives a base for each of the 21 words and, when it exists, an even base with the same word. The values B , k_B , and s_B refer to the base in the second column. The companion file `data/witnesses.csv` also contains the two tower residues modulo 10^{k_B+1} , which provide small exact certificates for the first phase. Later phases follow from the proved recurrence, not from extrapolation.

word	base a	B	v	k_B	s_B	even witness
2	9	1	1	1	2	48
4	11	2	1	3	4	36
5	5	4	2	10	5	—
6	52	2	1	1	6	52
8	2	3	1	1	8	2
9	101	2	2	6	9	—
19	901	2	2	6	1	—
28	18	3	2	2	2	18
46	3	2	1	1	4	12
64	16	2	1	3	6	16
82	39	1	1	1	8	88
2486	21	2	1	3	2	72
2684	7	2	2	2	2	—
3971	301	2	2	6	3	—
4268	6	3	1	4	4	6
4862	19	1	1	1	4	28
6248	4	2	1	1	6	4
6842	8	2	1	1	6	8
7931	701	2	2	6	7	—
8426	31	2	1	3	8	32
8624	77	1	1	1	8	—

For example, base 7 has $o = 4$, $r = 2$, $A = 1$, $t = 3$, $B = 2$, and $k_B = 2$. The residues $T_2 \equiv 543$ and $T_3 \equiv 343 \pmod{1000}$ give $s_B = 2$. Furthermore $C_5 = 3$, $u \equiv 1$, and $\lambda = 3$, so $2 \mapsto 6 \mapsto 8 \mapsto 4 \mapsto 2$. This independently yields 2684.

For base 21, $T_2 \equiv 4421$ and $T_3 \equiv 2421 \pmod{10000}$ give $s_2 = 2$, and $\lambda = 2$, hence 2486. The significance of this example for the currently listed OEIS values is discussed in Section 13.

9 The related 28-value conjecture

OEIS A376446 uses the prescribed anchor

$$H = v(a) + 2, \quad v(a) = \begin{cases} r, & 5 \nmid a, \\ t, & 5 \mid a, \end{cases} \quad (9.1)$$

instead of B [7]. Denote its reduced phase word by $\text{MPS}(a)$. The distinction is an actual rotation of a cycle, not a different notion of phase.

Lemma 9.1 (The prescribed anchor is safe). *For every admissible base, $B \leq H$.*

Proof. Even bases have $B \leq 3 \leq r + 2$. Bases ending in 5 have $B \leq t + 2$ by (5.2) and (4.6). For odd units with $t = r$, $B \leq 2$. If $t > r$, the increasing difference $A - \varepsilon r + b(t - r)$ is nonnegative by $b = \max(1, r - 1)$, so (5.6) gives $B \leq \max(2, r) \leq r + 2$.

If $t < r$ and $\varepsilon = -1$, then $a \equiv 3 \pmod{4}$, so $A = 1$ and the difference is $-r - 1 + b(r - t)$; it is nonnegative by $b = r + 1$. If $\varepsilon = 0$, it is $-A + b(r - t)$ with $A \leq t < r$, again nonnegative by $r + 1$. If $\varepsilon = 1$, it is already positive at $b = 0$, since $r > A$. Formula (5.6) finishes the proof. $\square$

Theorem 9.2 (The 28-value modular classification). *The exact range of MPS is*

$$\mathcal{E} \cup \{5, 9, 19, 91, 1397, 1793, 3179, 3971, 7139, 7931, 9317, 9713\}. \quad (9.2)$$

This is the 28-value set conjectured in OEIS A376446.

Proof. By Lemma 9.1, strict 2-adic and 5-adic regimes still give 5 and words in $\mathcal{E}$. For a balanced base $a = 1 + 10^v c$ we have $H = v + 2$ and

$$s_{H+j} \equiv -c^{v+3+j} \pmod{10}. \quad (9.3)$$

If $c \equiv 1$, this is always 9. If $c \equiv 9$, the parity of v gives 19 or 91. If $c \equiv 3$ or 7, the four residues of v modulo 4 give all four rotations of the corresponding four-cycle. These account for exactly the 11 odd unit words in (9.2).

All of them are attained by $a = 1 + c10^v$ with $c \in \{1, 3, 7, 9\}$ and $2 \leq v \leq 5$. The words of $\mathcal{E}$, in the order shown in (2.1), are attained by bases

$$9, 11, 52, 2, 33, 16, 3, 18, 4, 32, 6, 13, 19, 27, 7, 28,$$

respectively; $a = 5$ gives 5. These are exact finite modular evaluations, also recorded and checked in `data/modular_witnesses.csv`. $\square$

10 A density consequence

Corollary 10.1. *Each of the four sets*

$$\{a : \text{APS}(a) = 9\}, \quad \{a : \text{APS}(a) = 19\}, \quad \{a : \text{APS}(a) = 3971\}, \quad \{a : \text{APS}(a) = 7931\}$$

has natural density $1/900$ in the positive integers. Their union has density $1/225$.

Proof. Theorem 2.1 makes the balanced family necessary as well as sufficient. For a specified residue $c_0 \in \{1, 3, 7, 9\}$ and a specified $v \geq 2$, the relevant bases form the single arithmetic progression

$$a \equiv 1 + c_0 10^v \pmod{10^{v+1}},$$

with density $10^{-(v+1)}$. These progressions are disjoint as v varies, since $v = v_{10}(a - 1)$ is determined by a . The union over $v \geq N$ is contained in $a \equiv 1 \pmod{10^N}$, so its upper density is at most 10^{-N} . This bound justifies passage from finite sums to the infinite union. Therefore

$$\sum_{v=2}^{\infty} 10^{-(v+1)} = \frac{1}{900}$$

is the natural density for each specified word. Summing the four disjoint families gives $1/225$. $\square$

Thus the rarity of odd non-5 phase words has an exact arithmetic explanation. The finite test through base 100000 found 111 instances of each of these four words. This count is a check on the formulas; the density proof does not rely on it.

11 Unbounded onset and a sharp family

The phase alphabet is finite, but the minimal-speed onset is unbounded. A uniform search through a fixed number of heights is therefore not a proof of stabilization.

Proposition 11.1 (A constructive sharp-onset family). *For every integer $r \geq 3$, let a_r be the least positive representative of the Chinese remainder conditions*

$$a_r \equiv 2^{r-1} - 1 \pmod{2^r}, \quad (11.1)$$

$$a_r \equiv 2^{5^{r-1}} \pmod{5^{r+1}}. \quad (11.2)$$

Then

$$B(a_r) = r + 2, \quad v(a_r) = r - 1, \quad \text{APS}(a_r) = 5. \quad (11.3)$$

Hence the general sufficient bound $B \leq r + 2$ for odd units is sharp.

Proof. Condition (11.1) gives $A = 1$ and $t = r - 1$. The other condition gives $a_r \equiv 2 \pmod{5}$, so $o = 4$, and $a_r \equiv 3 \pmod{4}$, so $\varepsilon = -1$.

Put $n = 5^{r-1}$, which is odd. Applying Lemma 3.1 to $(-4)^n - 1 = -(4^n + 1)$ gives

$$v_5(4^{5^{r-1}} + 1) = v_5((-4) - 1) + (r - 1) = r.$$

The residue modulo 5^{r+1} in (11.2) therefore ensures $v_5(a_r^2 + 1) = r$ exactly. The two affine valuations are

$$x_b = 1 + b(r - 1), \quad y_b = (b - 1)r, \quad y_b - x_b = b - r - 1.$$

They are equal at $b = r + 1$, and 2-adic dominance starts strictly at $b = r + 2$. The increment into the tied height is still r ; the increment out is $r - 1$. Thus the first permanently constant speed is at $r + 2$, where all subsequent phases are 5. $\square$

For example $r = 3$ gives $a_3 = 1307$, $B = 5$, and speed 2. One tested member with $r = 32$ is

$$a_{32} = 311423787862411847003581666295807,$$

whose onset is 34 and whose speed is 31. At that onset $k_B = 1055$, so its first phase can be independently checked using residues modulo 10^{1056} , despite the inaccessible size of the actual tower. The bound itself is known in the congruence-speed literature [3]; the point here is the explicit CRT construction and the exact explanation of its crossing height.

12 Exact computation without constructing towers

12.1 The main algorithm

For a given admissible base, compute A, t when a is odd, and o, r when $5 \nmid a$. Propositions 4.1–4.3 and (5.1), (5.2), (5.6) give B, v , and k_B . If the 2-adic valuation is smaller, return 5. If they are equal at B , use the balanced formula. Otherwise compute a single phase s_B by modular tetration modulo 10^{k_B+1} and propagate it with the multiplier (6.1). Four direct phase evaluations are included as an independent internal check, rather than being needed by the mathematical algorithm.

A nonunit local valuation such as $v_2(a)T_{b-1}$ must not be constructed when it is enormous. To compare it with a known integer y , it suffices to compute $\min\{T_{b-1}, \lfloor y/v_2(a) \rfloor + 1\}$. This preserves the smaller valuation and detects equality exactly.

12.2 A reliable modular-tower routine

The supplied `Tower.mod` handles moduli $m = 2^u 5^w$. For each prime power $p^e \mid m$ there are two cases. If $p \nmid a$, reduce the exponent modulo $\varphi(p^e)$ and recurse on the lower-height tower. If $p \mid a$, compute the exponent only up to $\lceil e/v_p(a) \rceil$: at or above this threshold the residue is zero; below it the exponent is small and known exactly. Combine the two prime-power residues by the Chinese remainder theorem.

The recursion terminates because the height decreases. Every new modulus is again 2,5-smooth. The exponent reduction in the unit case is justified by Euler's theorem; in the nonunit case no unjustified exponent reduction is made.

The base cases are $T_0 = 1$ and $\min\{T_h, 1\} = 1$. For $h \geq 1$ and $M > 1$, to compute $\min\{T_h, M\}$, let q be the least nonnegative integer with $a^q \geq M$, compute $\min\{T_{h-1}, q\}$ recursively, and return M when the threshold is reached. Otherwise evaluate the small exact power. All of this uses integers, with no floating-point logarithms.

Corollary 12.1 (Polynomial-time computability). *The word $\text{APS}(a)$ can be computed in time polynomial in the bit length of a .*

Proof. Write L for that bit length. The valuations A, t, r are $O(L)$, since they are taken on $a \pm 1$ or $a^o - 1$ with $o \leq 4$. The onset bound gives $B = O(L)$ and the local profiles give $k_B = O(L^2)$. Thus the required modulus has $O(L^2)$ bits, not a tower-sized representation.

In a memoized modular recursion a state is specified by a height at most $B + 1$ and the two exponents of its smooth modulus. Those exponents remain $O(L^2)$; hence even the coarse bound on the number of possible states is polynomial in L . Each state uses polynomial-size integer arithmetic, modular exponentiation, and at most two recursive prime-power branches. The capped-tower calculations also use polynomial-size integers and bounded recursion depth. Standard multiplication and repeated squaring therefore suffice for a polynomial-time algorithm. No optimal complexity bound is claimed here. $\square$

12.3 A second algorithm used for checking

The verification program also implements full-modulus Euler reduction with exponent lifting. It computes $T_h \pmod{m}$ by reducing the exponent modulo $\varphi(m)$ and adding $\varphi(m)$ when the actual exponent is at least that large. If it is smaller, it uses the exact exponent instead.

For a prime power dividing m , an exponent at least $\varphi(m)$ is large enough to force the nonunit residue to zero, since $\varphi(m) \geq \varphi(p^e) \geq e$. In the unit factors, adding $\varphi(m)$ does not change the residue. This algorithm does not split the answer into prime-power residues and reconstruct it by CRT, so it exercises a different reduction path. It shares only the capped-tower helper, which is separately checked against directly constructed small towers.

13 Reproducible verification and source audit

13.1 What was actually checked

The included program was executed with Python 3.13.5, using standard-library integer arithmetic only. The saved run records:

Check	count or range
Admissible bases in $2 \leq a \leq 100000$	89999
Post-onset phase positions checked	719992
Direct-integer residue, cap, and valuation checks	3006
Independent Euler-algorithm comparisons	107
Structured sharp-onset examples	7
Largest structured onset	34
Distinct minimal-anchor words	21
Distinct even-base minimal-anchor words	14
Distinct prescribed-anchor words	28

For each sequentially tested base, the program checked eight post-onset phases, the exact increment at each tested height, and the failure of the limiting increment immediately before the proposed onset when $B > 1$. It also compared the four directly computed phases with the structural prediction from one phase and its multiplier, or with the balanced formula. The source code and its assertions make the scope of these tests explicit.

The sequential count excludes 1 and multiples of 10; it is therefore 89999, not 90000. The elapsed time in the saved successful run was 28.344 seconds on the execution environment used here. It is not a hardware-independent performance guarantee. The JSON report and CSV files contain the exact results, including the discrepancy described next.

13.2 The value currently listed at base 21

The accessed A376842 entry lists 6248 at base 21 [6]. With the height-one convention of A371074, the calculation is instead

$$A = t = 2, \quad r = 1, \quad \varepsilon = 1, \quad k_b = b + 1 \ (b \geq 1), \quad V_1 = 2, \quad B = 2. \quad (13.1)$$

The onset $B(21) = 2$ is independently the value listed in OEIS A372490 [9]. From the exact residues

$$T_2 \equiv 4421, \quad T_3 \equiv 2421 \pmod{10000}$$

and the multiplier $\lambda = 2$ follows

$$\text{APS}(21) = 2486. \quad (13.2)$$

The phase at height one is 6, so 6248 is the same cycle started one height earlier. It is therefore consistent with an anchoring slip rather than a failure of the periodicity or the 21-value range. The supplied checker records this mismatch explicitly; it does not silently modify the source data. Every other admissible listed value from base 2 through base 59 agrees with the calculation.

13.3 The height-zero ambiguity

Definitions phrased uniformly as a difference between the numbers of stable digits at heights b and $b - 1$ invite substituting $T_0 = 1$ at $b = 1$. The height-one convention recorded in A371074 instead counts the matching digits of a and a^a directly. This is the convention used throughout this paper, and it matches the examples $\text{APS}(301) = 3971$ and $\text{APS}(901) = 19$ in Section 3 of Ripà's 2025 paper [4].

For a balanced base, substituting $v_{10}(a - 1) = v$ as an initial count would make $B = 1$, not 2, and would give

$$9, \quad 1397, \quad 1793, \quad 91$$

for $c \equiv 1, 3, 7, 9$, respectively. These are different anchored invariants. Thus 901 is not a counterexample to the intended 21-value conjecture; it only exposes why the convention must be fixed. We resolve this ambiguity before stating the theorem rather than relying on an accidental rotation.

13.4 The base-5 length convention

Some discussions count only physically written digits at height two, assigning four stable digits to $5^5 = 3125$ even though its congruence with the next tower holds modulo 10^5 . Here we consistently use the exact modular count $k_2 = 5$. Either choice has $B(5) = 4$ and $\text{APS}(5) = 5$, so it does not affect the classification. It does affect the short initial list of speeds and should not be left implicit.

13.5 Literature status and limits of the claim

The target conjecture was checked in the original MathOverflow question, the 2025 arXiv PDF including its appendix, and the current OEIS entry. The PDF appendix matters: the HTML rendering consulted does not display that final appendix as completely as the PDF. The 2022 valuation paper was checked separately to distinguish existing speed results from the phase problem. A targeted search for the sequence identifier and phase-shift terminology located no subsequent proof of the complete classification.

This evidence supports selecting the problem as an explicitly outstanding conjecture in the consulted sources. It does not establish an exhaustive bibliographic negative or guarantee novelty of every lemma and corollary. No assertion of external peer review, Lean formalization, or Wolfram verification is made. The calculations reported here were performed locally in Python; the mathematical argument is supplied in full above.

14 Interpretation and further questions

The tower's size is largely irrelevant to its first changing decimal digit. The relevant mechanism is a comparison of two local divisibility profiles. When one profile is strictly smaller, normalization kills the contribution from the other prime. What remains is either the single residue 5 or a multiplicative orbit in the four-element group $\mathbb{F}_5^\times$. Exactly balanced profiles retain a decimal unit, and the anchor fixes which rotation of its orbit is seen.

The same reasoning explains why an arbitrary later height has the same first changing digit relative to *any* further tower. For $b \geq B$ and $h > b$,

$$T_h - T_b = \sum_{j=b}^{h-1} D_j \equiv D_b \pmod{10^{k_b+1}},$$

because every D_j with $j > b$ is divisible by 10^{k_b+v} and $v \geq 1$. Thus the phase really describes the first permanently unsettled digit, not merely a coincidental difference between two isolated heights.

There are natural extensions that this article does not assert as proved. One is a full density distribution over all 21 words, rather than the four balanced ones. Another is a comparable classification in other radices, where more local primes or nontrivial prime-power weights can change the finite-state dynamics. A third is a machine-checked formalization of the anchor lemma and the valuation-to-phase reduction. These are separate tasks; they are not gaps in the decimal classification proved here.

Conclusion. Under the documented height-one convention, the 21-value conjecture and its 14-value even-base refinement follow from exact local valuation formulas and a finite-group recurrence. The prescribed-height 28-value conjecture follows as a rotation classification. The most delicate step is not computing enormous towers: it is proving that the first constant speed occurs strictly after any isolated tie between the local valuations.

A Archive contents and reproduction

The archive contains the complete L^AT_EX source and compiled PDF, together with the following files:

File	purpose
code/phase_tetration.py	Exact smooth-modulus towers, valuations, onsets, phase words, and the sharp CRT family.
code/verify.py	Sequential, direct-integer, independent-reduction, structured, and published-data checks.
data/verification.json	Machine-readable results of the successful run.
data/verification.txt	Human-readable copy of that result.
data/witnesses.csv	All 21 witnesses, onsets, speeds, and exact first-phase residue certificates.
data/modular_witnesses.csv	Witnesses for all 28 prescribed-anchor words.
data/sharp_onsets.csv	Structured examples including onset 34.
README.md	Commands, conventions, and the limits of the computational checks.
SOURCE_STATUS.md	Sources, versions, and the initial-height/data audit.
build.sh, build.ps1	PDF build commands for common environments.

From the extracted directory, use Python 3.9 or later:

```
python code/phase_tetration.py 57
python code/phase_tetration.py 901
python code/verify.py --max-base 1000000
```

Do not use Python's `-O` option, which disables assertions. No Python packages beyond the standard library are required. The saved run used Python 3.13.5; the code targets Python 3.9+ syntax and APIs, but was not executed on every intervening interpreter version.

Build the document with a T_EX installation containing `newpxtext`, `newpxmath`, `tcolorbox`, and the usual AMS packages:

```
pdflatex -interaction=nonstopmode -halt-on-error \
  tetration_phase_classification.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  tetration_phase_classification.tex
```

The bibliography is self-contained in the source, so BibT_EX is not required. No font files or copies of third-party papers are included.

References

- [1] Marco Ripà, *On the constant congruence speed of tetration*, Notes on Number Theory and Discrete Mathematics **26**(3) (2020), 245–260. doi:10.7546/nntdm.2020.26.3.245-260.
- [2] Marco Ripà, *The congruence speed formula*, Notes on Number Theory and Discrete Mathematics **27**(4) (2021), 43–61. doi:10.7546/nntdm.2021.27.4.43-61.
- [3] Marco Ripà and Luca Onnis, *Number of stable digits of any integer tetration*, Notes on Number Theory and Discrete Mathematics **28**(3) (2022), 441–457. doi:10.7546/nntdm.2022.28.3.441-457. Consulted preprint: <https://arxiv.org/pdf/2210.07956>.
- [4] Marco Ripà, *Graham's number stable digits: An exact solution*, Notes on Number Theory and Discrete Mathematics **31**(3) (2025), 607–616. doi:10.7546/nntdm.2025.31.3.607-616. Consulted arXiv version 2, 12 September 2025: <https://arxiv.org/pdf/2411.00015v2>; see Definition 3.2 and the appendix on preprint page 9 for the target conjectures.
- [5] Marco Ripà, *A new Conjecture at OEIS sequence A376842*, MathOverflow question 481090, 22 October 2024. <https://mathoverflow.net/questions/481090/>. Accessed 19 September 2026; the retrieved question displayed no answer.
- [6] OEIS Foundation Inc., *A376842*, entry by Marco Ripà. <https://oeis.org/A376842/internal>. Accessed 19 September 2026; revision 39, 30 March 2026.
- [7] OEIS Foundation Inc., *A376446*, entry by Marco Ripà. <https://oeis.org/A376446/internal>. Accessed 19 September 2026; revision 40, 13 December 2025.
- [8] OEIS Foundation Inc., *A371074*, entry by Marco Ripà. <https://oeis.org/A371074>. Accessed 19 September 2026. The comments explicitly identify the total height-one matching-digit count with the height-one speed.
- [9] OEIS Foundation Inc., *A372490*, entry by Marco Ripà. <https://oeis.org/A372490>; table <https://oeis.org/A372490/b372490.txt>. Accessed 19 September 2026; used to cross-check the onset at base 21.